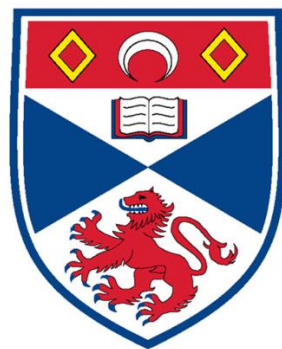


# Pre-project review for final-year project in Physics & Astronomy

Visualising the Abraham-Minkowski Light Field Momentum

**Matriculation No:** 200007312



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of  
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**Module:** PH5101 MPhys Physics

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# Visualising the Abraham-Minkowski Light Field Momentum

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## Introduction

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For over a century, there has been an ongoing debate concerning the form that the electromagnetic momentum density should take in dielectric media, known as the Abraham–Minkowski controversy. In recent years, the theory of optical momentum and energy transfer to matter has regained interest, much of which can be attributed to the growth in nanotechnology, where it finds practical applications [1]. In this review, we part from a revision of Maxwell’s equations and electromagnetic (EM) plane waves. We then derive Maxwell’s stress tensor (in vacuum) and identify the relation between the Poynting vector and the momentum density. Understanding this is crucial to extent the discussion to a spacetime context through the electromagnetic stress–energy tensor (EMSET) whose generalisation to dielectric media lies at the core of theoretical resolutions to the controversy. We review the main existing theories emphasising the form of the EMSET and the momentum density they produce, alongside their predicted momentum transfer. Finally, we lay out the methods and steps that will be utilised in this Final Year Project to produce and employ cubic dipole crystals to simulate light transfer in dielectric media to visualise and calculate momentum transfer to compare it with theoretical resolutions to the controversy.

## Background

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### Maxwell’s Equations and Plane Waves

Maxwell’s (microscopic) equations (hereon referred to as MEs) are defined as

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} \tag{1}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{2}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (3)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (4)$$

where  $\mathbf{E}$  and  $\mathbf{B}$  are the electric and magnetic fields respectively,  $\rho$  is the charge density,  $\mathbf{J}$  is the current density, and  $\mu_0$  and  $\epsilon_0$  are respectively the vacuum permeability and permittivity [2]. We use **bold unitalicized** text to denote that an object is a vector.

In empty space, both  $\rho$  and  $\mathbf{J}$  are equal to zero, simplifying Eq.1 and 4. Taking the curl of either, substituting the other and using a vector identity yields wave equations [2] of the form

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad (5)$$

$$\nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} \quad (6)$$

Here we identify the speed of light in vacuum

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \quad (7)$$

Generalising to the phase velocity of light  $v$  in any dielectric medium with relative permittivity  $\epsilon$  and permeability  $\mu$ ,

$$v = \frac{1}{\sqrt{\mu \epsilon}} \quad (8)$$

Which allows the definition of the refractive index  $n$  [3] as

$$n = \frac{c}{v} = \sqrt{\frac{\mu_0 \epsilon_0}{\mu \epsilon}} \quad (9)$$

The fact that the fields have zero divergence shows that the waves are purely transverse. We can thus employ plane wave solutions

$$\mathbf{E} = \mathbf{E}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (10)$$

$$\mathbf{B} = \mathbf{B}_0 e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (11)$$

Where  $\mathbf{k}$  is the wavevector,  $\omega$  is the angular frequency,  $\mathbf{E}_0$  and  $\mathbf{B}_0$  are the amplitude constants (pointing perpendicular to  $\mathbf{k}$  and each other),  $\mathbf{r}$  is the position vector and  $t$  is time. We can obtain the phase velocity from the dispersion relation  $\omega(k)$  [2]

$$v_\phi = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c \quad (12)$$

which we can extend to the case of dielectric media when we can approximate a linear dispersion. This allows a calculation of the refractive index of the medium.

### Maxwell's Stress Tensor and Poynting's Theorem

Our aim is to understand the transfer of momentum from EM waves to matter, in other words, how EM fields do mechanical work. Thus, we turn our attention to the Lorentz force

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = \int (\rho \mathbf{E} + \mathbf{J} \times \mathbf{B}) dV \quad (13)$$

where we have turned this into an integral over volume so we can talk about the force over charge distribution and not just point charges. The integrand is  $\mathbf{f}$  the force per unit volume. Substituting  $\rho$  and  $\mathbf{J}$  from Eqs.1 and 4 into Eq.13

$$\mathbf{f} = \epsilon_0 (\nabla \cdot \mathbf{E}) \mathbf{E} + \left( \frac{1}{\mu_0} \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \times \mathbf{B} \quad (14)$$

By employing the product rule with vector identities and further substituting Maxwell's equations, one can arrive at the following expression:

$$\begin{aligned} \mathbf{f} = \epsilon_0 [(\nabla \cdot \mathbf{E}) \mathbf{E} + (\mathbf{E} \cdot \nabla) \mathbf{E}] + \frac{1}{\mu_0} [(\nabla \cdot \mathbf{B}) \mathbf{B} + (\mathbf{B} \cdot \nabla) \mathbf{B}] \\ - \left[ \frac{1}{2} \epsilon_0 \nabla (|\mathbf{E}|^2) + \frac{1}{2\mu_0} \nabla (|\mathbf{B}|^2) \right] - \epsilon_0 \left[ \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{B}) \right] \end{aligned} \quad (15)$$

Identifying Poynting's vector

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \quad (16)$$

And introducing the Maxwell stress tensor [4]

$$\begin{aligned}\vec{\sigma} = \sigma_{ij} &= \epsilon_0 E_i E_j + \frac{1}{\mu_0} B_i B_j - \frac{1}{2} \delta_{ij} \left( \epsilon_0 E_{ij}^2 + \frac{1}{\mu_0} B_{ij}^2 \right) \\ &= \frac{1}{2} \left( \epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \vec{\mathbf{I}} - \epsilon_0 \mathbf{E} \otimes \mathbf{E} - \frac{1}{\mu_0} \mathbf{B} \otimes \mathbf{B}\end{aligned}\quad (17)$$

where  $i, j = x, y, z$  and  $\vec{\mathbf{I}}$  is the unit dyadic, we get [5]

$$\mathbf{f} = (\nabla \cdot \vec{\sigma}) - \frac{1}{c^2} \frac{\partial \mathbf{S}}{\partial t} \rightarrow \nabla \cdot \vec{\sigma} = \mathbf{f} + \frac{1}{c^2} \frac{\partial \mathbf{S}}{\partial t} \quad (18)$$

This should be reminiscent of Poynting's theorem [6,7]

$$\nabla \cdot \mathbf{S} = -\mathbf{E} \cdot \mathbf{J} - \frac{\partial}{\partial t} \frac{1}{2} \left( \epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) = -\frac{\partial u_{mech}}{\partial t} - \frac{\partial u_{EM}}{\partial t} \quad (19)$$

where  $\mathbf{E} \cdot \mathbf{J}$  gives the work density the field does on charges so we identify it as the rate of change of the mechanical energy density  $u_{mech}$ . Evidently,  $u_{EM}$  gives the electromagnetic energy density. By their structure, Eqs. 18 and 19 are continuity equations. Since  $\mathbf{f}$  is just the rate of change of the mechanical momentum density then

$$\mathbf{g} = \frac{\mathbf{S}}{c^2} \quad (20)$$

Is the electromagnetic momentum density.

So far, our discussion has taken vacuum as the medium for the EM fields. It is the correct generalisation of these equations to dielectric media that lies at the source of the Abraham–Minkowski controversy. In particular, it is the choice of the form of the momentum density around which much of the debate revolves.

### The Electromagnetic Stress–Energy Tensor

For any electromagnetic system, the dynamics must always satisfy the postulates of special relativity. In a Minkowski space (flat spacetime), it is convenient to rewrite Eqs. 18 and 19 as a single four-dimensional continuity equation because four-vectors are Lorentz-covariant [8]. Using the Einstein summation convention, this is

$$\eta^{\mu\sigma} f_{\sigma} = f^{\mu} = -\partial_{\nu} T^{\mu\nu} \quad (21)$$

where  $\mu, \nu = 0, 1, 2, 3$ , the metric

$$\eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1) \quad (22)$$

the “4D divergence”

$$\partial_\nu = \frac{\partial}{\partial x^\nu} = \left( \frac{1}{c} \frac{\partial}{\partial t}, \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) = \left( \frac{1}{c} \frac{\partial}{\partial t}, \nabla \right) \quad (23)$$

the four-vector combining the power and force densities

$$f_\nu = \left( \frac{1}{c} \frac{\partial u_{mech}}{\partial t}, f_x, f_y, f_z \right) = \left( \frac{1}{c} \frac{\partial u_{mech}}{\partial t}, \mathbf{f} \right) \quad (24)$$

and the electromagnetic stress-energy tensor (EMSET)

$$\vec{\mathbf{T}} = \begin{bmatrix} u_{EM} & S_x/c & S_x/c & S_x/c \\ c g_x & \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ c g_x & \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ c g_x & \sigma_{zx} & \sigma_{xx} & \sigma_{zz} \end{bmatrix} \quad (25)$$

$\sigma_{ij}$  are the elements of  $\vec{\sigma}$ . From Field Theory, it is known that, to conserve angular momentum, the total stress-energy tensor from a closed system must be symmetric [9]. As well, this tensor must be form-invariant under Lorentz transformations [8]. Note that only the total stress-energy tensor of the matter-field system must be symmetric and invariant [10], so there is choice regarding the matter counterpart that makes the total tensor mathematically appropriate. The physical significance of this choice depends on how the matter-field interaction is described [8].

## Competing Theories

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### Minkowski

In 1908, Minkowski (subscript M) developed a stress tensor focusing on the symmetry requirements from Lorentz transformations [11] considering linear and isotropic media without distinguishing between bound and free charges and currents [12]. In his formulation, he uses the MEs from above but with a different version of Eq.4

$$\nabla \times \mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \quad (26)$$

where  $\mathbf{H} = \frac{\mathbf{B}}{\mu}$  is the auxiliary field,  $\mathbf{D} = \epsilon \mathbf{E}$  is the electric displacement and  $\mathbf{J}_f$  is the free charge density. He makes no distinction between the field and matter contributions as the vectors  $\mathbf{E}$ ,  $\mathbf{H}$ ,  $\mathbf{D}$ , and  $\mathbf{B}$  contain the matter response through the constitutive relations with  $\mathbf{P}$  and  $\mathbf{M}$  as the polarisation and magnetisation. Hence, any description of the propagation of energy and momentum will include some response from the medium thus including the dynamics of the material [1].

His tensor

$$\overleftrightarrow{\sigma}_{\mathbf{M}} = \frac{1}{2}(\mathbf{D} \cdot \mathbf{E} + \mathbf{B} \cdot \mathbf{H})\overleftrightarrow{\mathbf{I}} - \mathbf{D} \otimes \mathbf{E} - \mathbf{B} \otimes \mathbf{H} \quad (27)$$

with momentum density

$$\mathbf{g}_{\mathbf{M}} = \mathbf{D} \times \mathbf{B} \quad (28)$$

Taking the divergence operator of this tensor along with using the MEs and vector calculus properties, we find the associated electromagnetic force density

$$\mathbf{f}_{\mathbf{M}} = -\frac{1}{2}|\mathbf{E}|^2 \nabla \epsilon - \frac{1}{2}|\mathbf{H}|^2 \nabla \mu \quad (29)$$

where the refractive index is assumed to be time independent.

Applying a similar method for obtaining the continuity equations above, we get

$$\mathbf{S}_{\mathbf{M}} = \mathbf{E} \times \mathbf{H} \quad (30)$$

$$u_{EM\mathbf{M}} = \frac{1}{2}(\mathbf{D} \cdot \mathbf{E} + \mathbf{B} \cdot \mathbf{H}) \quad (31)$$

$$\frac{\partial u_{mech}}{\partial t}_{\mathbf{M}} = \mathbf{E} \cdot \mathbf{J}_f \quad (32)$$

Minkowski's EMSET is relativistically invariant, but not symmetric, as  $\mathbf{S}_{\mathbf{M}} \neq \frac{\mathbf{g}_{\mathbf{M}}}{c^2}$  [13].

Using a quasi-stationary analysis and considering a plane EM wave incident on a slab of linear transparent material, in terms of the wave-particle duality of a photon, the Minkowski momentum can be calculated as [1]

$$p_{\mathbf{M}} = \hbar k = n \frac{\hbar \omega}{c} \quad (33)$$

This is related to the reduced wavelength in matter (as expected from wave refraction) resulting in a linear relation with  $n$ . Here  $p$  is momentum and  $\hbar$  is the reduced Planck's constant. This formulation seems to agree with most of the experiments so far [8]. It is suggested that this tensor corresponds to the canonical momentum of the electromagnetic wave in dielectric media and is responsible for translations within or with respect to a medium [14].

## Abraham

In 1909, Abraham (subscript Ab) developed the same tensor as Minkowski's, but used the momentum density [15]

$$\mathbf{g}_{Ab} = \frac{1}{c^2} \mathbf{E} \times \mathbf{H} \quad (34)$$

The force density

$$\begin{aligned} \mathbf{f}_{Ab} &= -\frac{1}{2} |\mathbf{E}|^2 \nabla \varepsilon - \frac{1}{2} |\mathbf{H}|^2 \nabla \mu + \frac{n^2 - 1}{c^2} \frac{\partial}{\partial t} (\mathbf{E} \times \mathbf{H}) \\ &= \mathbf{f}_M + \frac{\partial}{\partial t} (\mathbf{g}_M - \mathbf{g}_{Ab}) \end{aligned} \quad (35)$$

The energy continuity equation matches Minkowski's, so  $\frac{\partial u_{mech}}{\partial t}_M = \frac{\partial u_{mech}}{\partial t}_{Ab}$ ,  $u_{EM_{Ab}} = u_{EM_M}$  and  $\mathbf{S}_{Ab} = \mathbf{S}_M$ . Abraham's EMSET is symmetric, but not invariant [13]. Using the same analysis as above, the Abraham's momentum

$$p_{Ab} = mv = \frac{1}{n} \frac{\hbar \omega}{c} \quad (36)$$

is related to Einstein's mass equivalent  $m = \frac{\hbar \omega}{c^2}$  and is proportional to  $1/n$  [1]. It is suggested that this tensor corresponds to the kinetic momentum and is responsible for the overall centre-of-mass translations of a medium [14].

Ampère

This is the most common one in modern EM theory textbooks (subscript A). This formulation employs these versions of Eqs. 1 and 4

$$\nabla \cdot \mathbf{E} = \frac{\rho_f}{\varepsilon_0} - \frac{1}{\varepsilon_0} \nabla \cdot \mathbf{P} \quad (37)$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}_f + \mu_0 \frac{\partial \mathbf{P}}{\partial t} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \nabla \times \mathbf{M} \quad (38)$$

where  $\rho_f$  is the free charge density. It represents macroscopic media via an equivalent current density

$$\mathbf{J}_{eb} = \mathbf{J}_f + \frac{\partial \mathbf{P}}{\partial t} + \nabla \times \mathbf{M} \quad (39)$$

and charge density

$$\rho_{eb} = \rho_f - \nabla \cdot \mathbf{P} \quad (40)$$

producing the magnetisation and polarisation. The force density is given by the Lorentz force law (Eq. 14), thus the force

$$\mathbf{f}_A = -\nabla \cdot \overleftrightarrow{\mathbf{\sigma}}_A - \frac{\partial \mathbf{g}_A}{\partial t} \quad (41)$$

with

$$\overleftrightarrow{\mathbf{\sigma}}_A = \frac{1}{2} \left( \epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \overleftrightarrow{\mathbf{I}} - \epsilon_0 \mathbf{E} \otimes \mathbf{E} - \frac{1}{\mu_0} \mathbf{B} \otimes \mathbf{B} \quad (42)$$

$$\mathbf{g}_A = \epsilon_0 \mathbf{E} \times \mathbf{B} \quad (43)$$

From the continuity equations,

$$\mathbf{S}_A = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B} \quad (44)$$

$$u_{EMA} = \frac{1}{2} \left( \epsilon_0 |\mathbf{E}|^2 + \frac{1}{\mu_0} |\mathbf{B}|^2 \right) \quad (45)$$

$$\frac{\partial u_{mech}}{\partial t} \bigg|_A = \mathbf{E} \cdot \mathbf{J}_f + \mathbf{E} \cdot \frac{\partial \mathbf{P}}{\partial t} + \mathbf{E} \cdot (\nabla \times \mathbf{M}) \quad (46)$$

The Ampère EMSET is symmetric and relativistically invariant [13]. Using the same analysis as above [1]

$$p_A = \frac{\mu}{\mu_0} \frac{1}{n} \frac{\hbar \omega}{c} \quad (47)$$

There is a subtle additional contribution to the momentum – known as “hidden momentum” – arising from the treatment of magnetisation as an electric current effect as the system even though microscopically at rest contains internally moving parts [8]

$$p_h = \frac{1}{c^2} (\mathbf{m} \times \mathbf{E}) \quad (48)$$

where  $\mathbf{m}$  is the magnetic dipole moment. This generates an extra force density  $\mathbf{f}_h$  and  $\mathbf{S}_h$  such that  $\mathbf{S}_{total} = \mathbf{S}_A + \mathbf{S}_h = \mathbf{E} \times \mathbf{H} = \mathbf{S}_{Ab/M}$ .

Chu

It (subscript C) uses the E-H representation, where the presence of matter is added as source terms in Maxwell's equation [8]. The material response is represented by equivalent electric current density  $\mathbf{J}_e$ , magnetic current density  $\mathbf{J}_m$ , electric charge density  $\rho_e$ , and magnetic charge density  $\rho_m$ . In stationary media [1],

$$\epsilon_0 \nabla \cdot \mathbf{E} = \rho - \nabla \cdot \mathbf{P} = \rho_e \quad (49)$$

$$\mu_0 \nabla \cdot \mathbf{H} = -\mu_0 \nabla \cdot \mathbf{M} = \rho_m \quad (50)$$

$$\nabla \times \mathbf{E} + \mu_0 \frac{\partial \mathbf{H}}{\partial t} = -\mu_0 \frac{\partial \mathbf{M}}{\partial t} = -\mathbf{J}_m \quad (51)$$

$$\nabla \times \mathbf{H} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \frac{\partial \mathbf{P}}{\partial t} + \mathbf{J} = \mathbf{J}_e \quad (52)$$

The electromagnetic and matter-field parts are separated. The EM part is identical to the Maxwell stress-energy tensor  $\overleftrightarrow{\sigma}_c = \overleftrightarrow{\sigma}$  with momentum density  $\mathbf{g}_c = \mathbf{g}_{Ab}$  noting that the fields must be calculated according to the E-H equations.  $\mathbf{S}_c = \mathbf{S}_{Ab}$ ,  $u_{EMc} = u_{EMA}$  and

$$\frac{\partial u_{mech}}{\partial t}_c = \mathbf{E} \cdot \mathbf{J}_e + \mathbf{H} \cdot \mathbf{J}_m \quad (53)$$

The Chu stress tensor  $\overleftrightarrow{\sigma}$  is symmetric and invariant [13]. The interaction part is described by the tensor

$$\overleftrightarrow{\sigma}_{int} = -\mathbf{P} \otimes \mathbf{E} - \mu_0 \mathbf{M} \otimes \mathbf{H} \quad (54)$$

with  $\mathbf{g}_{int} = \mathbf{0}$  obtained either with a thermodynamic procedure or with a volumetric force balance argument.

The momentum matches the Abraham one [1].

## The Project

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Over a century of theory and experiments have expanded the understanding of electrodynamics yet left the controversy unresolved. In such a context, computational simulations offer a new avenue for evaluating solutions. Our plan is to simulate the interaction of light travelling in a dielectric medium with a solid body inside the medium and measure the momentum it imparts.

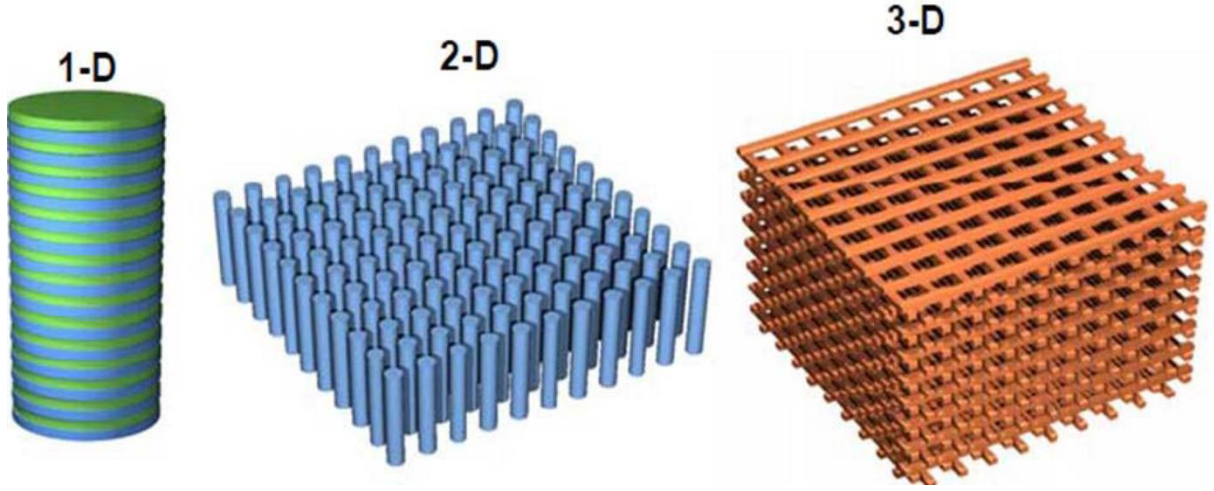


Figure 1: Schematic illustrations of photonic crystals with simple periodic patterns. Taken from [16].

When discussing electromagnetic fields in dielectric media in terms such as the magnetisation, polarisation, and refractive index we are actually considering the average effect of the medium-field interaction and we are treating the medium as if it were a homogenous structure. This picture ignores the reality the media are not homogenous and that between the particles that compose a medium there is vacuum. Hence, we aim to exploit this by dehomogenising the medium by modelling it as a photonic crystal employing the coupled dipole method. Photonic crystals are optical periodic nanostructures (Fig.1) designed to affect the motion of photons in a similar way to how the periodic potential in a semiconductor crystal affects the motion of electrons by defining allowed and forbidden electronic energy bands [16].

The coupled dipole method (Fig. 2) is an analytic framework in which each sublattice of the crystal is modelled as a point dipole – comprising both electric and magnetic moments. This enables a simple and efficient simulation and calculation of the bulk and boundary dispersion of 1D and 2D photonic crystals [17].

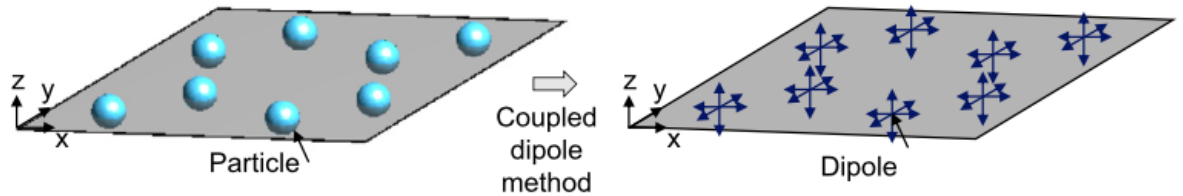


Figure 2: Schematic of the coupled dipole method. Taken from [17].

We plan to set up a 1D crystal with fixed spacing  $L$  and dipoles have moments

$$\mathbf{p}_i = \alpha_i \mathbf{E} \quad (55)$$

where the dipole moment coefficients  $\alpha_i$  can be complex incorporating a term for the phase shift between the dipole and the incoming field, and  $\mathbf{E}$  is the electric component of a plane EM wave.  $L$  should be such that when Fourier transforming the crystal structure we get a crystal  $\mathbf{k}$ -vector much larger than the field wavevector (i.e.  $\lambda$  of the light is much larger than the crystal periodicity – this is the quasi-static regime) to ensure that  $|\mathbf{k}| \ll$  the Brillouin zone edge and we can approximate a linear dispersion facilitating the calculation of the refractive index. This should correspond to the refractive index when the medium is homogenised.

We plan to start with a 1D interface made up of two contiguous lattices with similar spacing but dissimilar dipole moment coefficients. In the crystal dipole model, calculating the momentum transfer is straightforward and unambiguous because we set the integration surface always in the vacuum between dipoles where the EMSET is well defined and understood. Combining our results for the momentum and the refractive index we can assess the accuracy of the theoretical predictions for the form of the light-field momentum discussed in the previous section. If time allows, we would like to extend this treatment to a 2D interface.

## Conclusion

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We have derived the electrodynamic theory that gives rise to the Abraham-Minkowski controversy when we try to generalise the EMSET to dielectric media. We have then covered 4 of the main theories which attempt to resolve this debate describing the different formulations they propose for separating the matter and field parts of the system. Lastly, we have proposed a plan for calculating and visualising the light field momentum by simulating photonic crystals using the coupled dipole method.

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